

Test Final Report

Project: HTU Capstone (Sauce Demo & DummyJSON)

Author: Zaid Alabed

Date: February 6, 2026

1) Executive Summary

The quality assurance phase for the Swag Labs e-commerce platform and the DummyJSON API has been completed.

The project involved Manual Testing, API Automation, Performance Testing, and UI Automation and the results indicate that the **Backend (API)** is stable and production ready.

However, the **Frontend (UI)** contains critical blockers in the checkout flow, which is in the problem_user interface, Then the recommendation is to proceed with the API deployment but **hold the UI deployment** until the critical defects are resolved.

2) System Under Test + Assumptions

Systems Tested:

- **Frontend:** SauceDemo Focusing on user journey and UI responsiveness.
- **Backend:** DummyJSON Focusing on data integrity and load handling.

Assumptions:

- All testing was performed on the Microsoft Edge browser (latest version).
- The standard_user was used to verify happy paths, while the problem_user was used for negative testing and defect discovery.
- The test environment and internet connection remained stable throughout the execution.

3) Test Strategy and Scope

- **Strategy:** A hybrid approach was adopted, utilizing manual testing for exploratory purposes and UI quirks and automation for regression stability.
- **In-Scope:** Login functionality, Inventory Management, Cart operations, Checkout Flow (Steps 1 & 2) and API CRUD operations.
- **Out-of-Scope:** Payment Gateway processing and direct database validation.

4) Manual Testing

- **Scenarios Covered:** Over 20 End-to-End test cases were executed.
- **Execution Summary:**
 - **Passed:** 12 cases (Happy Paths).
 - **Failed:** 8 cases (Negative Paths/Defects).
- **Coverage:** 100% of the core functional requirements were executed, you can go to Docs/Manual Test Cases.xlsx as a reference for detailed steps.

5) Defects

- **Total Defects Found:** 8
- **Top Issues:**
 1. **Critical:** Checkout Blocked – The "Last Name" field is disabled for the problem_user, preventing the transaction.
 2. **High:** Inventory – Product images fail to load (broken links).
- **Severity Distribution:** Critical and High severity issues account for 62.5% of total defects.
- **Reference:** See Docs/Bug reports saucedemo.xlsx for full logs.

6) API Testing (Postman)

- **Collection Design:** The collection is structured to test CRUD operations (Create, Read, Update, Delete) on Products and Auth endpoints.
- **Key Assertions:**
 - Status Code is 200 or 201.
 - Response Time is less than 500ms.
 - JSON Schema Validation is correct.
 - Response Body contains expected data fields.
- **Newman Report Summary:** The suite was executed via the CI/CD pipeline and achieved a 100% pass rate.

7) Performance Testing (k6)

- **Test Plan:** Stress testing the DummyJSON API to verify stability under concurrent load.

- **Load Profile:**
 - **Virtual Users:** 15 VUs.
 - **Duration:** 30 Seconds.
 - **Ramp-up:** 0 seconds (Instant load).
- **Results & KPIs:**
 - **Error Rate:** 0.00% (Pass).
 - **Average Response Time:** 180ms.
 - **p95 Response Time:** 323ms (Target < 2000ms).
 - **Throughput:** 27 requests per second.
- **Analysis:** The API demonstrated excellent stability with zero timeouts or server errors.

8) UI Automation

- **Approach:** Scripts were built using Selenium to automate the Golden Path (Login -> Add to Cart -> Checkout).
- **Structure:** Scripts are organized in the UI_Automation folder by test scenario.
- **Sample Results:**
 - Login_Test: Pass
 - Cart_Addition_Test: Pass
 - Checkout_Flow_Test: Pass